



Report

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Summary

① Tasks Status

② Learned Concepts

③ Selected Article

Tasks Status

Task	Status	
2026-06 RETINHA presentation		Link
2026-06 Scientific report		Link
Journal Article		Link
2026-12 Scientific report		Link
Dissertation		Link

Tasks Status

Task	Status	Link
Model correlated uncertainty in Scaled Spectra		Notebook
↔ Check uncertainty prediction in Atlas experiment ins1759506		
↔ Check effect on pt scaled spectra (Thiago's work)		

Learned Concepts

Viscosity opposite of Collective Behavior [1]	✓
Holographic Theorem predicted viscosity that resembles QGP $\frac{\eta}{s}$ (AdS/CFT) [1]	✓
Initial conditions and Shear viscosity competes in generating the predicted outcomes in heavy ion collision models [1]	✓

Recent studies

Articles	Status	Link
Bayesian estimation of the specific shear and bulk viscosity of quark–gluon plasma	✓	Bernhard, Nature, 2019

Thesis	Status	Link
Bayesian parameter estimation for relativistic heavy-ion collisions	✓	Bernhard, PhD, 2018

Selected Article

VarP-GP: cost-efficient Bayesian emulation of quark-gluon plasma modeling with variable statistical precision

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We present VarP-GP, a new cost-efficient Bayesian emulator for expensive computational models with variable statistical precision. We focus on the interpretation of measurements of the quark-gluon plasma (QGP) generated in high-energy nuclear collisions, through comparison to numerical models using Bayesian Inference. Such inference calculations are computationally expensive and require surrogate model emulation, which is commonly implemented using Machine Learning (ML)-based Gaussian processes (GPs). Emulator training data are generated by Monte Carlo simulations whose numerical precision depends on the computational resources utilized; improved precision entails greater computational cost. This study utilizes JETSCAPE simulations of inclusive hadron and jet measurements in nuclear collisions at RHIC and the LHC. The VarP-GP emulator combines information from multiple simulation runs with varying precision across the model parameter space, taking advantage of the smoothness in that space of QCD-driven processes. Comparison to a traditional emulator approach shows a marked reduction in emulator uncertainty at fixed computational cost, indicating that knowledge of the overall contours of the parameter design space is more important for precise emulation than detailed information at a more limited number of design points. As an initial application of VarP-GP, a computationally-expensive model parameter sensitivity study of jet quenching data is reported. The VarP-GP emulator enables new multi-model and many-observable calibrations of QGP data and modeling, which would otherwise not be possible with achievable computing resources.

Figure

References I

- [1] Jonah E. Bernhard. “Bayesian parameter estimation for relativistic heavy-ion collisions”. en. arXiv:1804.06469 [nucl-th]. PhD thesis. arXiv, Apr. 2018. URL: <http://arxiv.org/abs/1804.06469> (visited on 12/12/2025).